W.H. Croxford High School · Mathematics Department

Mathematics 20-1 Course Outline

2026–27
5 credits

Teacher	Email	Google Classroom	Class website
Mr. Ryan Cathcart Room 2997	**rcathcart@rvschools.ab.ca** Students and parents welcome	**Code given in class** Website link, files, announcements	**rmcathcart-lab .github.io/math20-1** Lessons, practice, calendar, tests

About the course

Math 20-1 is the second course on the pre-calculus (−1) route. It builds the algebra and functions that Math 30-1 and calculus lean on: sequences and series, radicals, trigonometry, factoring and quadratic functions and equations, rational expressions, absolute value and reciprocal functions, and systems of equations and inequalities.

Math 10C → **Math 20-1** → Math 30-1 (+ Math 31)

Math 20-1 follows Math 10C and is worth 5 credits. It is a 20-level math, which a high school diploma needs (100 credits in all).

Success in math is mostly persistence. Everyone in this room can do well in this course by doing the practice, asking questions early and learning from mistakes. Wrong answers are where the learning happens.

A typical class

1. **Arrive ready**: laptop charged, calculator, pencil and paper on your desk.
2. **Warm-up** on the board while attendance is taken.
3. **Lesson**: I teach from the class website. Take notes on paper, or print the notes booklet for the unit.
4. **Practice** on the website: new numbers every time, hints when you're stuck, and your level on each outcome.
5. **Exit slip** almost every day: a short check on today's outcome, written on the website.

Classes run about 88 minutes, bell to bell. The class calendar on the website shows each day's lesson and every test date, and it is always up to date.

The course at a glance

Unit	Topic	Lessons	Outcomes
1	Sequences and Series	9	RF9 · RF10
2	Radical Expressions	7	AN2
3	Trigonometry	9	T1 · T2 · T3
4	Factoring, Quadratic Equations and Radical Equations	8	RF1 · RF5 · AN3
5	Quadratic Functions and Equations	11	RF3 · RF4 · RF5
6	Rational Expressions and Equations	10	AN4 · AN5 · AN6
7	Absolute Value and Reciprocal Functions	8	AN1 · RF2 · RF11
8	Systems of Equations and Inequalities	8	RF6 · RF7 · RF8

70 lessons and a unit test at the end of every unit. Dates for your class are on the class calendar.

What you will learn: the 20 outcomes

From Alberta Education's Mathematics 10–12 Program of Studies (2008). These outcomes are what you are assessed on.

Algebra and Number

AN1 Demonstrate an understanding of the absolute value of real numbers.

AN2 Solve problems that involve operations on radicals and radical expressions with numerical and variable radicands.

AN3 Solve problems that involve radical equations (limited to square roots).

AN4 Determine equivalent forms of rational expressions (limited to numerators and denominators that are monomials, binomials or trinomials).

AN5 Perform operations on rational expressions (limited to numerators and denominators that are monomials, binomials or trinomials).

AN6 Solve problems that involve rational equations (limited to numerators and denominators that are monomials, binomials or trinomials).

Trigonometry

T1 Demonstrate an understanding of angles in standard position [0° to 360°].

T2 Solve problems, using the three primary trigonometric ratios for angles from 0° to 360° in standard position.

T3 Solve problems, using the cosine law and sine law, including the ambiguous case.

Relations and Functions

RF1 Factor polynomial expressions of the form: $ax^2 + bx + c$, $a \neq 0$; $a^2x^2 - b^2y^2$, $a \neq 0$, $b \neq 0$; $a(f(x))^2 + b(f(x)) + c$, $a \neq 0$; $a^2(f(x))^2 - b^2(g(y))^2$, $a \neq 0$, $b \neq 0$ where a, b and c are rational numbers.

RF2 Graph and analyze absolute value functions (limited to linear and quadratic functions) to solve problems.

RF3 Analyze quadratic functions of the form $y = a(x - p)^2 + q$ and determine the: vertex; domain and range; direction of opening; axis of symmetry; x- and y-intercepts.

RF4 Analyze quadratic functions of the form $y = ax^2 + bx + c$ to identify characteristics of the corresponding graph, including: vertex; domain and range; direction of opening; axis of symmetry; x- and y-intercepts and to solve problems.

RF5 Solve problems that involve quadratic equations.

RF6 Solve, algebraically and graphically, problems that involve systems of linear-quadratic and quadratic-quadratic equations in two variables.

RF7 Solve problems that involve linear and quadratic inequalities in two variables.

RF8 Solve problems that involve quadratic inequalities in one variable.

RF9 Analyze arithmetic sequences and series to solve problems.

RF10 Analyze geometric sequences and series to solve problems.

RF11 Graph and analyze reciprocal functions (limited to the reciprocal of linear and quadratic functions).

How you are assessed

Math 20-1 uses **outcomes-based assessment**. Your mark shows which outcomes you can do and how well, not how many points you collected. For each outcome you earn an **indicator**. On a unit test, it reads straight from three boxes of questions:

Beginning box	Progressing box	Mastery box
the core skill	the skill in typical questions	harder, multi-step and new situations

A box is **secure** when you earn ¾ of its points and **half done** at ½:

MAS	Mastery	Beginning, Progressing and Mastery boxes all secure.	**85–100%**
ADV	Advancing	Beginning and Progressing secure; at least half of Mastery.	**75–85%**
PRG	Progressing	Beginning and Progressing secure.	**65–75%**
EMG	Emerging	Beginning secure; at least half of Progressing.	**55–65%**
BEG	Beginning	Beginning box secure.	**45–55%**
LIM	Limited	At least half of the Beginning box.	**25–45%**
NM	Not Meeting	Less than half of the Beginning box.	**0–25%**
IE	Insufficient Evidence	Not enough work yet to judge, such as a missed test not yet made up.	—

From outcomes to your report card

Part marks and unit grades

Questions can earn part marks: a small slip keeps ⅔ of the point, and a bigger but reasonable mistake keeps ⅓. Your **unit grade** is the average of the unit's outcomes. Exit slips check one or two outcomes from the day's lesson, and they are graded the same way as a unit test: by their Beginning, Progressing and Mastery boxes.

Your final mark

Your report card mark comes from your grades on the 20 outcomes. Everything you do **in class** adds evidence for the outcomes it covers: unit tests, exit slips and practice in class. When I make a judgment call on an outcome, I use what I've seen you do in class.

Tests and exit slips

Unit tests and exit slips are written **on the class website**, in class, on your laptop. They pop up full screen when I launch them, and every student gets their own numbers. Tests end at the end of class. **Leaving the test window, switching tabs or leaving full screen submits your work and locks the test**; only I can unlock it.

Once I release the results, you see your answers, the correct answers and a worked solution for each question.

Missed a test?

Make it up at the **Test Centre** in the Learning Commons, Fridays 2:15–3:45. Fill out the missed-assessment form first. If you miss your Test Centre time, you lose that chance.

Practice and due dates

Each lesson's practice is on the website, with a due date on the class calendar. Due dates set the pace, so you're ready for the exit slip and the unit test. Practice isn't marked down for being late, since marks come from work in class, but unfinished practice matters for retests.

Retests: earn another try

A retest is earned with practice, and your **best result counts**, so a retest can only help.

1. **Do the practice.** Finish every required question in the unit without seeing the solution. (Saw one? Redo a new version without it.)
2. **Request a retest.** The button appears on that unit on your home page.
3. **Write it in class.** I open a retest just for you, with new numbers, while the class works.

Want another retest after that? Finish **all the extra practice** in the unit first.

> **Coins.** Every outcome has a coin beside your name on the website, from Blank (Not Meeting) up to Proof (Mastery). It shows the best level you have reached from practice in class or a unit test.

What to bring every day

- **Laptop or Chromebook**, charged. Croxford is a bring-your-own-device school; ask the office about borrowing a Chromebook for the year.
- **Graphing calculator**: a TI-84 Plus (or Plus CE) or a Casio fx-9750GIII. Not your phone. Both are also built into the class website.
- **Pencil, eraser** and **paper or a notebook** for notes (or your printed notes booklet).

Calculators

Calculators are allowed on **almost everything** in Math 20-1; a no-calculator question says so. **CAS** calculators (ones that do algebra with letters, such as the TI-Nspire CAS, TI-89 or Casio ClassPad) are not allowed. On tests I may ask you to clear stored notes or programs. Not sure about yours? Bring it in and we'll check it.

Expectations

- **Be on time and ready** to start when the bell goes.
- **Phones off and away** during class (the provincial rule). If you leave the room, your phone stays with me; take the hall pass, one person at a time.
- **Compassionate Cavs:** be kind, encourage each other, and listen when someone is teaching. This is a room where you can be wrong out loud.
- **Own your learning.** Missed a class? The lesson, notes and practice are on the website: catch up before the next class and ask me about anything unclear.

Getting help

Asking early is the biggest predictor of doing well. Ask during work time in class (that's what it's for), use the hints in the practice, or come see me at lunch or before or after school. The only time I won't help is during a test.

Academic integrity

Every assessment must show **your own** thinking. On tests and exit slips, any of these is cheating:

- talking to or communicating with anyone, by any means;
- using notes, devices, websites or apps that weren't allowed;
- using **AI** or photo-solver apps for any part of the work;
- sharing questions or answers with students who haven't written yet.

On practice, copying someone's work costs you the practice. AI and solver apps are fine for *checking* your work, not for producing it.

When academic dishonesty happens, I will talk with you and contact home. It can mean a 0 on that assessment, or an alternate assessment of the outcome at my discretion. Serious or repeated cases are referred to administration.

For parents and guardians

Marks are posted in **PowerSchool** by outcome: an indicator for each outcome a unit test or exit slip has covered, plus each unit grade.

For the day-to-day picture, ask your student to show you their home page on the class website: their level on every outcome, their coins, what's due and the next test date.

This course follows Rocky View Schools **AP 360**, Assessment and Communication of Student Learning, and **AP 223**, Student Reassessment (rockyview.ab.ca), and the Croxford student handbook. Questions about this outline? Email rcathcart@rvschools.ab.ca.